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Bundle Protocol Contact Probability Block
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Abstract

This document defines the Contact Probability Block (CPB), a Bundle Protocol Version 7 extension block for carrying bundle-conditioned forwarding-success probabilities. Each entry identifies a decision node, a candidate next hop, and the estimated probability that the enclosing bundle will reach its destination before expiration if that forwarding action is selected.

CPB transports the output of a routing estimator; it does not define an estimator or a forwarding algorithm. Its common probability meaning permits estimates derived from different routing systems to be represented without standardizing algorithm-specific ranks or utilities. The compact CBOR format uses deterministic IEEE 754 binary16 probabilities and existing BPV7 security mechanisms. This document is Experimental.

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Table of Contents

1. Introduction	2
1.1. Requirements Language	3
1.2. Terminology	3
2. Probability Semantics	3
3. Block Format	4
3.1. CDDL	4
3.2. Fields	5
3.3. Deterministic Encoding	6
3.4. Block Processing Flags and CRC	6
4. Test Vector	7
5. Processing	7
5.1. Production	8
5.2. Consumption and Selection	8
5.3. Malformed and Unsupported Blocks	9
6. Relationship to Routing and Advertisement Systems	9
7. Fragmentation and BPsec	9
8. Security Considerations	10
9. IANA Considerations	10
10. Implementation Status	11
11. References	11
11.1. Normative References	11
11.2. Informative References	12
Appendix A. Acknowledgements	12
Author's Address	12

1. Introduction

A DTN routing implementation can have information that is specific to one bundle and one forwarding decision. Examples include an estimate conditioned on the bundle's destination, expiration time, size, priority, and a route computation performed at a particular time. Such an estimate is not equivalent to a general advertisement about a neighbor or scheduled contact.

This document defines a BPv7 extension block, the Contact Probability Block (CPB), that carries those estimates with the bundle. A path computation element, bundle source, or forwarding node can produce a

CPB. A routing implementation can consume it as one input to a local forwarding decision. Nodes that do not implement CPB continue to process the bundle according to [RFC9171].

The name "Contact Probability Block" is retained for continuity with earlier work. The standardized value is deliberately narrower than a generic contact-quality score: it is the probability of timely end-to-end delivery conditional on a specific forwarding action.

1.1. Requirements Language

The key words "MUST", "MUST NOT", "REQUIRED", "SHALL", "SHALL NOT", "SHOULD", "SHOULD NOT", "RECOMMENDED", "NOT RECOMMENDED", "MAY", and "OPTIONAL" in this document are to be interpreted as described in BCP 14 [RFC2119] [RFC8174] when, and only when, they appear in all capitals, as shown here.

1.2. Terminology

Decision node The BP node at which the represented forwarding choice is made.

Candidate next hop The BP node to which the decision node can attempt to forward the bundle.

Forwarding-success probability The estimated probability that the enclosing bundle reaches its destination before its lifetime expires, conditional on the decision node selecting the candidate next hop.

Producer The node or path computation element that generated a CPB.

2. Probability Semantics

Every probability in a CPB has the following meaning:

| P(the enclosing bundle is delivered to its destination before
| bundle expiration | the identified decision node selects the
| identified next hop at the evaluation time)

The event includes the attempted transfer to the next hop and all subsequent forwarding. It therefore incorporates, as applicable, contact availability, transmission success, downstream routing, capacity, and the remaining bundle lifetime. The estimator determines how those factors are modeled; this specification does not.

Values are probabilities in the closed interval $[0, 1]$, not ranks, costs, confidence intervals, or arbitrary normalized utilities. A producer **MUST NOT** encode an algorithm-specific value unless it has been converted into the meaning above. In particular, raw MaxProp costs and RAPID utilities are not CPB probabilities.

A consumer **MUST** compare only entries whose decision-node EID matches the consumer's Node ID using the applicable EID-scheme comparison rules. The consumer then matches candidate-next-hop EIDs in the same way. For the ipn URI scheme, comparison is performed on decoded EID information, not CBOR byte equality, as required by [RFC9758].

CPB does not prescribe a forwarding rule. A consumer **MAY** rank eligible candidates by probability, combine a probability with latency or resource policy, or decline to use CPB. A consumer **MUST NOT** interpret absence of an entry as probability zero.

3. Block Format

CPB is a BPv7 canonical extension block. Its block-type-specific data is a CBOR byte string containing one deterministic CBOR data item. The data item conforms to the CDDL [RFC8610] below.

3.1. CDDL

```

<CODE BEGINS>
cpb-btsd = bstr .cbor cpb-data

cpb-data = {
  0: entries,
  1: evaluation-time,
  2: validity-duration,
  3: producer-node,
  * (uint .ge 4) => any
}

entries = [1*8 forwarding-entry]

forwarding-entry = [
  decision-node: eid,
  candidate-next-hop: eid,
  forwarding-success-probability: probability
]

evaluation-time = uint
validity-duration = uint .gt 0
producer-node = eid
probability = float16

eid = $eid .within eid-structure
eid-structure = [uri-code: uint, scheme-specific-part: any]

; The dtn and ipn sockets below summarize the EID forms used by
; this specification. RFC 9171 and RFC 9758 are authoritative.
$eid /= [1, tstr / 0]
$eid /= [2, ipn-ssp2 / ipn-ssp3]
ipn-ssp2 = [fqnn: uint, service-number: uint]
ipn-ssp3 = [
  allocator-identifier: uint .lt 4294967296,
  node-number: uint .lt 4294967296,
  service-number: uint
]
<CODE ENDS>

```

Figure 1: CPB Block-Type-Specific Data

An implementation MUST apply the complete EID syntax and processing rules from [RFC9171] and updates such as [RFC9758]. The open CDDL socket permits BP URI schemes registered after this document.

3.2. Fields

entries (key 0) One through eight forwarding entries. A CPB MUST

NOT contain duplicate pairs of decision-node and candidate-next-hop EIDs after scheme-defined normalization.

evaluation-time (key 1) DTN time, in milliseconds since the DTN epoch, at which the probabilities were evaluated. This uses the DTN time basis from [RFC9171].

validity-duration (key 2) Positive duration in milliseconds beginning at evaluation-time. It is not a decay constant and does not extend the bundle lifetime.

producer-node (key 3) Full BP EID identifying the asserted producer. This field alone does not authenticate that identity.

extension fields A consumer MUST ignore unrecognized unsigned-integer keys after validating the four required fields. An extension MUST NOT change the semantics of a base field.

The three EID positions identify BP nodes. The null endpoint dtn:none MUST NOT appear in any of them.

3.3. Deterministic Encoding

Producers MUST encode cpb-data using the deterministic CBOR requirements of [RFC8949], including preferred serialization and ascending map-key order.

Every probability MUST use the three-octet CBOR binary16 representation. A producer converts its internal value to IEEE 754 binary16 using roundTiesToEven after verifying that the input is finite and in [0, 1]. Positive and negative zero MUST be encoded as positive zero. NaN, infinity, and values outside [0, 1] are malformed input. A positive input smaller than half the least positive binary16 subnormal rounds to zero. Other subnormal binary16 results are valid and MUST NOT be flushed to zero by the wire encoder.

A consumer MUST reject a CPB from routing use if a probability uses a CBOR integer, binary32, binary64, or a non-deterministic encoding. It MUST NOT clamp a malformed value into the valid interval.

3.4. Block Processing Flags and CRC

A CPB producer MUST set all block processing control flags to zero. In particular, CPB is not replicated into every fragment. This avoids presenting an estimate calculated for an unfragmented bundle as if it had been calculated for each fragment.

CRC type and value are governed by [RFC9171] and [RFC9173]. This document does not replace those rules. In particular, use of CRC type zero depends on an applicable integrity service, and removal of the final integrity service can require a forwarding node to add a CRC before forwarding.

4. Test Vector

The following diagnostic notation contains two forwarding actions for decision node ipn:200.0. The producer is ipn:50.0, the evaluation time is 16203904, and the validity duration is 3600000 milliseconds.

```
<CODE BEGINS>
{
  0: [
    [[2, [200, 0]], [2, [100, 0]], 0.75],
    [[2, [200, 0]], [1, "//relay.example/"], 0.5]
  ],
  1: 16203904,
  2: 3600000,
  3: [2, [50, 0]]
}
<CODE ENDS>
```

Figure 2: CPB Diagnostic Form

The 67-octet deterministic cpb-data encoding is:

```
A400828382028218C800820282186400F93A008382028218C8008201702F2F
72656C61792E6578616D706C652FF93800011A00F74080021A0036EE800382
0282183200
```

Figure 3: CPB Data Hexadecimal Encoding

Wrapping that item as block-type-specific data adds the byte-string header 5843. A complete 75-octet test block using private block type 200, block number 2, flags 0x00, and CRC type zero begins 8518C80200005843 followed by the cpb-data bytes above. CRC type zero is conforming only when permitted by the integrity rules in Section 3.4.

5. Processing

5.1. Production

A producer MUST evaluate probabilities for the exact enclosing bundle and populate the evaluation time, validity duration, and producer EID. It MUST NOT copy a CPB to a different bundle unless the estimates were evaluated for that bundle as well.

A bundle MUST NOT contain more than four CPBs or more than 1024 octets of aggregate CPB block-type-specific data. A producer that would exceed either limit MUST omit an older CPB or decline to add the new one. These limits bound append-only growth and parsing work.

A node MUST NOT modify a CPB covered by a BIB. It can append a new CPB under its own producer identity, subject to the limits above.

5.2. Consumption and Selection

Before routing use, a consumer performs the following steps:

1. Parse and validate the deterministic CBOR and all required fields.
2. Ignore the CPB if local DTN time is outside the half-open interval [evaluation-time, evaluation-time + validity-duration), allowing only a locally configured clock tolerance.
3. Apply local authentication and authorization policy to the producer. Trust eligibility and trust rank are evaluated before freshness.
4. Retain only entries whose decision-node matches the local Node ID.

If multiple eligible CPBs remain, a consumer MUST select one complete CPB rather than combine entries from different producers. It first selects the locally highest trust rank, then the greatest evaluation-time, then the lowest canonical block number as a deterministic tie breaker. A deployment can define trust ranks, but freshness MUST NOT override a higher trust rank.

A receiver presented with more than four CPBs or more than 1024 aggregate CPB octets MUST exclude the excess from routing use in ascending canonical block-number order. It MAY apply a stricter local resource limit.

5.3. Malformed and Unsupported Blocks

A node MUST NOT use a malformed CPB for routing. It SHOULD retain the block unchanged when forwarding the bundle so that BIB target relationships are not damaged. Any block deletion, bundle deletion, or status reporting follows the block processing flags and error procedures of [RFC9171] and BPSec processing in [RFC9172].

Failure to parse or authorize CPB metadata does not by itself make the payload undeliverable. The node continues using routing information available through other mechanisms.

6. Relationship to Routing and Advertisement Systems

CPB standardizes a transport representation, not an estimator. A PROPHET implementation [RFC6693], a contact-plan uncertainty model, or another system can produce a CPB only after it calculates the common forwarding-success probability in Section 2. Algorithm names and local formulas are intentionally absent from the wire format.

SAND advertisements [I-D.ietf-dtn-bp-sand] distribute neighborhood and routing information independently of a particular data bundle. CPB carries the result of a computation conditioned on the enclosing bundle. A producer can use SAND-advertised schedule, delay, rate, or error information as inputs to that computation; it does not copy those raw measurements into CPB.

For example, two bundles at the same node can have the same destination but different sizes and expiration times. A short high-rate contact can yield a high forwarding-success probability for the small bundle and a low probability for the large bundle. A neighborhood advertisement describes the contact once; CPB can carry the two different bundle-conditioned results.

7. Fragmentation and BPSec

A CPB is evaluated for the complete enclosing bundle. Fragmentation changes the object, its size, and potentially its remaining delivery conditions. A consumer MUST NOT use a CPB found in a fragment for routing. A source that requires CPB-assisted routing throughout the path SHOULD set the primary block's "bundle must not be fragmented" flag.

Fragmentation also changes primary-block information and can invalidate a BIB whose integrity scope includes that information. A bundle source that applies a BIB to a CPB MUST also set the primary block's "bundle must not be fragmented" flag. A forwarding node MUST NOT depend on a BIB-protected CPB remaining usable after

fragmentation. A deployment requiring both integrity-protected CPB and fragmentation SHOULD use encapsulation or another profile that explicitly defines compatible security processing.

This conservative rule avoids depending on unresolved general extension-block fragmentation behavior. It does not alter the fragmentation or BPSec procedures in [RFC9171], [RFC9172], or [RFC9173].

8. Security Considerations

A false probability can divert traffic, waste scarce contacts, or suppress a better route. CPB is therefore untrusted routing input unless authenticated and authorized by local policy. By default, an implementation MUST NOT use an unauthenticated CPB to override a route selected from authenticated local information.

A BIB can provide integrity and data-origin authentication. When a security context identifies a BIB security source, the asserted producer-node MUST match the identity authorized for that security source; otherwise the CPB is ineligible for routing use. A BIB is a message authentication mechanism and is not necessarily a public key signature.

Freshness checks limit replay but do not replace authentication. A producer should choose the shortest validity duration compatible with the estimator and expected propagation delay. Implementations MUST perform checked arithmetic when adding evaluation time and validity duration.

BCB confidentiality can make a CPB unavailable to intermediate routers. A producer MUST NOT encrypt CPB when it expects nodes outside the BCB security destination to consume it.

The entry, block-count, and aggregate-size limits in Section 5 reduce CPU, memory, and bandwidth amplification. Implementations should also rate-limit diagnostics for malformed or unauthorized CPBs.

9. IANA Considerations

This document requests assignment of one value from the "Bundle Block Types" registry in the "Bundle Protocol" registry group.

Value	Description	Reference
TBD1	Contact Probability Block	This document

Table 1: Requested Bundle Block Type

The value 200 is used only by the reference implementation and test vectors because it is in the experimental/private-use range. It MUST NOT be used as the permanent interoperable value. This document creates no metric-type registry.

10. Implementation Status

This section is to be removed before publishing as an RFC.

This section records implementation information in the format encouraged by [RFC7942] and is to be removed before publication as an RFC.

The author maintains a Python reference encoder and decoder at <https://github.com/luckyseoul/draft-perry-dtn-cpb>. It implements deterministic binary16 encoding, EID-preserving entries, complete-field validation, BTSD wrapping, and byte-exact conformance vectors.

11. References

11.1. Normative References

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Appendix A. Acknowledgements

Brian Sipos provided early review comments on the CDDL, embedded CBOR, fragmentation, EID handling, timestamps, and IANA considerations. Those comments materially improved this work. Remaining errors are the author's.

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DTN CPB

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